

EXERCISE 17.4.1

Suppose that a queuing system has 2 servers, an exponential interarrival time of 2 hours and exponential service time with mean of 2 hours. A customer arrives at 12.00 noon.

1. What is the probability that the next arrival will be

- before 1.00 pm
- between 1.00 pm and 2.00 pm
- after 2.00 pm

2. Suppose no additional customers arrive before 1.00 pm,

what is the probability that the next arrival will come between 1.00 pm and 2.00 pm?

3. What is the probability that the number of arrivals

between 1.00 pm and 2.00 pm will be

- 0
- 1
- 2 or more?

4. Suppose that both servers are serving customers at 1.00 pm.

What is the probability that neither customer will have the service completed

- before 2 pm
- before 1.10 pm
- before 1.01 pm

• INTERARRIVAL TIME: 2 hours $\Rightarrow \exp(\frac{1}{2})$

• SERVICE TIME: 2 hours $\Rightarrow \exp(\frac{1}{2})$

1. First arrival within first hour: $IP(X \leq 1) = 1 - e^{-\frac{1}{2} \cdot 1} = 1 - e^{-\frac{1}{2}}$

First arrival between 1pm and 2pm:

$$IP(1 \leq X \leq 2) = e^{-\frac{1}{2} \cdot 1} - e^{-\frac{1}{2} \cdot 2} = e^{-\frac{1}{2}} - e^{-1}$$

First arrival after 2pm: $IP(X \geq 2) = e^{-\frac{1}{2} \cdot 2} = e^{-1}$

2. First arrival between 1pm and 2pm given no arrivals before 1pm:

$$IP(X \leq 2 \mid X \geq 1) = IP(X \leq 1) = 1 - e^{-\frac{1}{2}} \quad (\text{LACK OF MEMORY})$$

3. IP (no arrivals between 1pm and 2pm)

$$= IP(\text{no arrivals in 1 hour}) = IP(X \geq 1) = e^{-\frac{1}{2}}$$

$$IP(1 \text{ arrival in 1 hour}) = \frac{1}{2} e^{-\frac{1}{2}} \quad (\text{POISSON})$$

$$IP(2 \text{ or more arrivals in 1 hour}) =$$

$$1 - IP(0 \text{ or 1 arrival in 1 hour}) = 1 - \frac{3}{2} e^{-\frac{1}{2}}$$

4. 2 servers working, each at a rate of 1 x 2 hours

$\Rightarrow$ together they complete 1 service per 1 hour

$$\Rightarrow Y \sim \exp(1)$$

'before 2pm' = 'within 1 hour' $IP(Y > 1) = e^{-1}$

1.10

10 min

$$IP(Y > \frac{1}{6}) = e^{-\frac{1}{6}}$$

1.01

1 min

$$IP(Y > \frac{1}{60}) = e^{-\frac{1}{60}}$$

EXERCISE 17.4-3

The time required by a mechanic to repair a machine has an exponential distribution with a mean of 4 hours.

However, a special tool would reduce this mean to 2 hours.

If the mechanic repairs the machine is less than 2 hours

he is paid 100€ otherwise he is paid 80€.

Determine the mechanic expected increase in pay per machine if he uses the special tool.

WITHOUT TOOL : repair time $X \sim \exp(\frac{1}{4})$

WITH TOOL : repair time $Y \sim \exp(\frac{1}{2})$

$$\begin{aligned}\text{EXPECTED PAY WITHOUT TOOL} &= 100 \cdot \mathbb{P}[X \leq 2] + 80 \cdot \mathbb{P}[X > 2] \\ &= 100(1 - e^{-\frac{1}{2}}) + 80e^{-\frac{1}{2}} \\ &= 100 - 20e^{-\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\text{EXPECTED PAY WITH TOOL} &= 100 \cdot \mathbb{P}[Y \leq 2] + 80 \cdot \mathbb{P}[Y > 2] \\ &= 100(1 - e^{-1}) + 80e^{-1} \\ &= 100 - 20e^{-1}\end{aligned}$$

$$\text{DIFFERENCE: } 20e^{-\frac{1}{2}} - 20e^{-1} \approx 6.77$$

A queuing system has 3 servers with expected service times of 30 minutes, 20 min and 15 min. All service times are exp dist. Each server has been busy with a ^{current} customer for 10 min. Determine the expected waiting time until one of these customers completes the service.

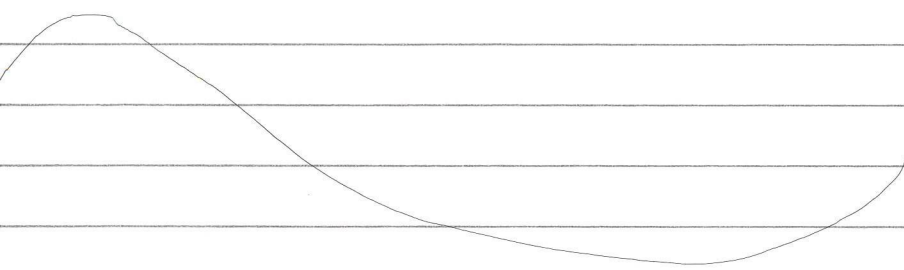
3 servers with 30 20 15 min service time, serve
 respectively 2 3 4 customers per hour
 together the serve 9 customers per hour

$\Rightarrow$ when they are all busy it takes $1/9$ hour for one of the servers to complete the service i.e. $60/9$ min ≈ 6.6 min

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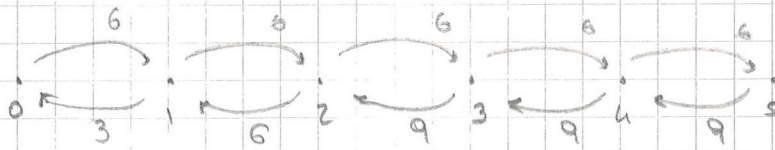
For an oral exam, students arrive according to a Poisson distribution with an average of 6 students per hour. There are 3 examiners available for the examination. To each examiner, it takes on average 20 minutes to evaluate a student. If all the examiners are busy, the arriving new students have to wait for their turn. Since the room is small, if there are more than two students waiting, the new arriving students have to come back later. Therefore, there are at most 5 students in the system: 3 taking the exam and 2 waiting.

1. What is the expected number of students in the system at any point in time?
2. Suppose the examiners are ready to start the exam and there are no students waiting yet. How long does it take before a student arrives and have to wait to take the exam?



a.

a.



$$\text{RATE IN} = \text{RATE OUT}$$

$$\begin{cases} 3\pi_1 = 6\pi_0 \\ 6\pi_2 = 6\pi_1 \\ 9\pi_3 = 6\pi_2 \\ 9\pi_4 = 6\pi_3 \\ 9\pi_5 = 6\pi_4 \\ \pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 + \pi_5 = 1 \end{cases}$$

$$\pi_1 = 2\pi_0$$

$$\pi_2 = \pi_1 = 2\pi_0$$

$$\pi_3 = \frac{6}{9}\pi_2 = \frac{4}{3}\pi_0$$

$$\pi_4 = \frac{6}{9}\pi_3 = \frac{8}{9}\pi_0$$

$$\pi_5 = \frac{6}{9}\pi_4 = \frac{16}{27}\pi_0$$

$$\pi_0 + 2\pi_0 + 2\pi_0 + \frac{4}{3}\pi_0 + \frac{8}{9}\pi_0 + \frac{16}{27}\pi_0 = 1$$

$$(27 + 54 + 54 + 36 + 24 + 16)\pi_0 = 27$$

$$\pi = (27, 54, 54, 36, 24, 16) \cdot \frac{1}{211}$$

$$\begin{aligned} \# \text{ STUDENTS} &= 0 \cdot \frac{27}{211} + 1 \cdot \frac{54}{211} + 2 \cdot \frac{54}{211} + 3 \cdot \frac{36}{211} + 4 \cdot \frac{24}{211} + 5 \cdot \frac{16}{211} \\ &= \frac{446}{211} \sim 2.11 \end{aligned}$$

b.

Ex time to go from x to 4

↓
limit state with someone waitingGOAL: $E_0 = ?$ (0 is the beginning of the exam)

$$\begin{cases} E_0 = \frac{1}{6} + E_1 \\ E_1 = \frac{1}{9} + \frac{3}{9}E_0 + \frac{6}{9}E_2 \\ E_2 = \frac{1}{12} + \frac{6}{12}E_1 + \frac{6}{12}E_3 \\ E_3 = \frac{1}{15} + \frac{9}{15}E_2 + \frac{6}{15}E_4 \\ E_4 = 0 \end{cases}$$

$$E_0 = 1.625 \text{ hours}$$

$$\begin{cases} E_0 = 10 + E_1 \\ E_1 = \frac{20}{3} + \frac{1}{3}E_0 + \frac{2}{3}E_2 \\ E_2 = 5 + \frac{1}{2}E_1 + \frac{1}{2}E_3 \\ E_3 = 4 + \frac{3}{5}E_2 \\ E_4 = 0 \end{cases}$$

$$E_0 = 92.5 \text{ minutes}$$

$$E_2 = \frac{1}{12} + \frac{1}{2} E_1 + \frac{1}{2} \left(\frac{1}{15} + \frac{3}{5} E_2 \right)$$

$$= \frac{7}{60} + \frac{1}{2} E_1 + \frac{3}{10} E_2$$

$$E_2 = \left(\frac{7}{60} + \frac{1}{2} E_1 \right) \frac{10}{7} = \frac{1}{6} + \frac{5}{7} E_1$$

$$E_1 = \frac{1}{9} + \frac{1}{3} E_0 + \frac{2}{3} \left(\frac{1}{6} + \frac{5}{7} E_1 \right)$$

$$= \frac{2}{9} + \frac{1}{3} E_0 + \frac{10}{21} E_1$$

$$E_1 = \left(\frac{2}{9} + \frac{1}{3} E_0 \right) \frac{21}{11} = \frac{42}{99} + \frac{7}{11} E_0$$

$$E_0 = \frac{1}{6} + \frac{42}{99} + \frac{7}{11} E_0$$

$$E_0 = \frac{117}{198} \cdot \frac{11}{4} = \frac{1287}{792} = 1.625 \text{ hours}$$